

5a	$\text{Current} = E / (R+r)$ $= 2.0 / (7.5+0.5)$ $= 0.25 \text{ A}$
bi	$I = Anvq$ $R = \rho L/A \rightarrow A = (6.9 \times 10^{-7}) (1.2) / 7.5 = 1.104 \times 10^{-7}$ $v = (0.25) / [(1.104 \times 10^{-7}) (8.0 \times 10^{28}) (1.6 \times 10^{-19})]$ $= 1.77 \times 10^{-4}$ $= 1.8 \times 10^{-4} \text{ m s}^{-1}$
ii	Since volume remains unchanged, stretching the wire will reduce the cross sectional area but increasing the length since $v [= \frac{I}{Anq} = \frac{V}{RAnq} = \frac{V}{\frac{\rho L}{A}(Anq)} = \frac{V}{\rho L nq}]$ is inversely proportional to L, drift velocity decreases.
ci	$V_{xJ} = 0.25(7.5) \times (0.90/1.20)$ $= 1.4 \text{ V}$

ii	Since there is no current flowing through the cell B at balanced length, the presence of the resistance in the galvanometer will have no effect on the balanced length
iii	The additional resistor is added in series with cell A or parallel with resistance wire in the primary circuit.
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